

QUBO Formulation: Subset Sum Problem

Subset Sum Problem

Problem Statement. Given a set of N positive integers $\{a_1, a_2, \dots, a_N\}$ and a target value T , the Subset Sum Problem seeks a subset of these integers whose sum is as close as possible to T . Formally, the task is to select a subset that minimizes the absolute deviation between the sum of the selected elements and the target value.

Decision Variables. For each integer a_i with index $i \in \{1, \dots, N\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if the i -th integer is included in the chosen subset, and 0 otherwise. The total number of binary variables is $n = N$.

QUBO Derivation. Step 1. The original objective is to minimize the absolute deviation:

$$\min_{x \in \{0,1\}^N} \left| \sum_{i=1}^N a_i x_i - T \right|. \quad (1)$$

Since the absolute value is non-negative, minimizing the absolute deviation is equivalent to minimizing its square. We therefore define the unconstrained objective function:

$$\min_{x \in \{0,1\}^N} \left(\sum_{i=1}^N a_i x_i - T \right)^2. \quad (2)$$

Step 2. The problem specification contains no constraints. The feasible region comprises all 2^N binary assignments, so no constraint equations need to be written.

Step 3. With no constraints, no penalty terms are constructed. The Hamiltonian consists solely of the expanded objective function.

Step 4. We expand the squared objective to obtain the QUBO Hamiltonian $H(x)$:

$$H(x) = \left(\sum_{i=1}^N a_i x_i - T \right)^2 \quad (3)$$

$$= \sum_{i=1}^N a_i^2 x_i^2 - 2T \sum_{i=1}^N a_i x_i + T^2 + \sum_{i \neq j} a_i a_j x_i x_j. \quad (4)$$

Applying the idempotent property of binary variables, $x_i^2 = x_i$, the linear terms simplify. The quadratic cross-terms can be grouped over unique pairs $i < j$:

$$H(x) = \sum_{i=1}^N (a_i^2 - 2Ta_i)x_i + 2 \sum_{1 \leq i < j \leq N} a_i a_j x_i x_j + T^2. \quad (5)$$

Step 5. Comparing $H(x)$ to the standard QUBO form $H(x) = \sum_i Q_{ii}x_i + \sum_{i < j} Q_{ij}x_i x_j + c$, we read off the general closed-form expressions for the Q matrix entries and the constant offset:

$$Q_{ii} = a_i^2 - 2Ta_i, \quad i \in \{1, \dots, N\}, \quad (6)$$

$$Q_{ij} = 2a_i a_j, \quad 1 \leq i < j \leq N, \quad (7)$$

$$c = T^2. \quad (8)$$

Penalty Weight Justification. The problem formulation contains no constraints; therefore, the feasible region is the entire hypercube $\{0, 1\}^N$. Consequently, no penalty terms or penalty weights are required. The Hamiltonian is composed entirely of the expanded objective function and the constant offset c . If constraints were introduced in a generalized variant, a penalty weight λ would be chosen such that $\lambda > \Delta_{\max}$, where $\Delta_{\max}$ is the maximum possible reduction in the objective value achievable by violating a single constraint, ensuring that infeasible states always carry higher energy than any feasible state.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^N$ is mathematically equivalent to minimizing the squared deviation $(\sum_{i=1}^N a_i x_i - T)^2$. Because the square function is strictly monotonic for non-negative arguments, the binary assignment that minimizes the squared deviation also minimizes the absolute deviation. The constant offset $c = T^2$ shifts all energy levels uniformly and does not affect the location of the minimum. Since the problem has no constraints, every binary assignment is feasible, and the global minimum of $H(x)$ corresponds exactly to the subset whose sum is closest to T .

Illustrative Example. Consider a concrete instance with $N = 3$ input integers $\{a_1, a_2, a_3\} = \{2, 3, 5\}$ and target value $T = 6$. Substituting these values into the general formulas yields the following concrete QUBO coefficients:

$$Q_{11} = 2^2 - 2(6)(2) = -20, \quad Q_{22} = 3^2 - 2(6)(3) = -27, \quad Q_{33} = 5^2 - 2(6)(5) = -35, \quad (9)$$

$$Q_{12} = 2(2)(3) = 12, \quad Q_{13} = 2(2)(5) = 20, \quad Q_{23} = 2(3)(5) = 30, \quad (10)$$

$$c = 6^2 = 36. \quad (11)$$

The resulting Hamiltonian is:

$$H(x) = -20x_1 - 27x_2 - 35x_3 + 12x_1x_2 + 20x_1x_3 + 30x_2x_3 + 36. \quad (12)$$

Evaluating $H(x)$ over all $2^3 = 8$ binary assignments reveals that the minimum energy is $H_{\min} = 1$. This optimal energy is achieved by the subsets corresponding to $x = (0, 0, 1)$, $x =$

$(1, 1, 0)$, and $x = (1, 0, 1)$, which represent the subsets $\{5\}$, $\{2, 3\}$, and $\{2, 5\}$, respectively. All three subsets yield a sum of 5 or 7, deviating from the target $T = 6$ by exactly 1, which is the minimum possible deviation for this instance.